

Lecture 17: Double Integrals over General Regions

Background & Motivation

Last time we integrated over rectangles. Most regions of interest—disks, triangles, regions between curves—aren't rectangular. Today we handle general regions using Type I and Type II setups, and introduce polar coordinates for circular symmetry. The Gaussian integral, which cannot be evaluated by single-variable methods, yields beautifully to the polar approach.

1. Warm-Up: Double Integral over a Rectangle

Evaluate $\int_0^2 \int_1^3 (3xy^2) dy dx$.

1. Inner integral (y): $\int_1^3 3xy^2 dy = 3x \left[\frac{y^3}{3} \right]_1^3 = x(27 - 1) = 26x$
2. Outer integral (x): $\int_0^2 26x dx = [13x^2]_0^2 = 52$
3. Result: 52

2. Type I Regions (Vertical Slicing)

Definition 1: Type I Region

$$R = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

- Integrate y first (inner), then x (outer)
- Think: vertical lines sweep left to right across the region

3. Type II Regions (Horizontal Slicing)

Definition 2: Type II Region

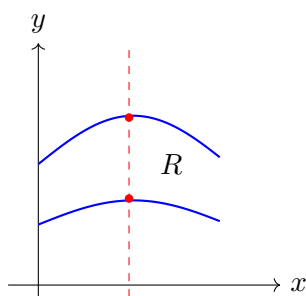
$$R = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

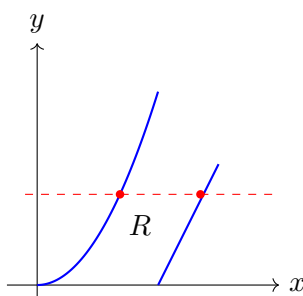
- Integrate x first (inner), then y (outer)
- Think: horizontal lines sweep bottom to top across the region

How to Classify a Region

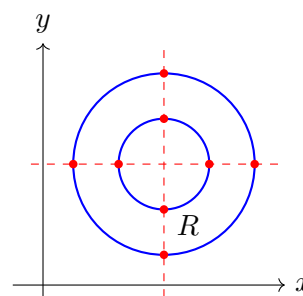
- Sketch the region and find all intersection points
- Type I – can every vertical line ($x = c$) enter through one curve and exit through one curve?
- Type II – can every horizontal line ($y = c$) enter through one curve and exit through one curve?
- Some regions work as both; pick whichever is simpler
- If neither works cleanly, **split** the region



Type I Region



Type II Region



Neither Type

A Type I region is bounded above and below by functions of x — every vertical line (red) hits the boundary exactly twice. Type II is analogous with horizontal lines. An annulus is *neither*: both horizontal and vertical slice lines can cross the boundary four times, so it must be split.

4. Examples

Example 1: Type I: Region Between Two Parabolas

Evaluate $\iint_R (x + 2y) dA$ where R is bounded by $y = x^2$ and $y = 2 - x^2$.

1. **Sketch and intersections:** $x^2 = 2 - x^2 \implies 2x^2 = 2 \implies x = \pm 1$.

Lower curve: $y = x^2$. Upper curve: $y = 2 - x^2$.

2. **Set up (Type I):**

$$\int_{-1}^1 \int_{x^2}^{2-x^2} (x + 2y) dy dx$$

3. **Inner integral (y):** $[xy + y^2]_{x^2}^{2-x^2} = x(2 - x^2) + (2 - x^2)^2 - x \cdot x^2 - x^4$
 $= 2x - x^3 + 4 - 4x^2 + x^4 - x^3 - x^4 = 2x - 2x^3 - 4x^2 + 4$

4. **Outer integral (x):** Note $2x - 2x^3$ is odd on $[-1, 1]$ so integrates to 0.

$$\int_{-1}^1 (-4x^2 + 4) dx = 2 \int_0^1 (-4x^2 + 4) dx = 2 \left[-\frac{4x^3}{3} + 4x \right]_0^1 = 2 \left(\frac{8}{3} \right)$$

5. **Result:** $\boxed{\frac{16}{3}}$

Example 2: Polar: Integral over a Disk

Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$.

1. **Convert to polar:** $x^2 + y^2 = r^2$, $dA = r dr d\theta$.

Disk: $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

2. **Set up:** $\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} d\theta \cdot \int_0^2 r^3 dr$

3. **Evaluate:** $= 2\pi \cdot \left[\frac{r^4}{4}\right]_0^2 = 2\pi \cdot 4$

4. **Result:** $\boxed{8\pi}$

5. Polar Coordinates for Double Integrals

Double Integrals in Polar

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

– The extra factor of r comes from the area element: $dA = r dr d\theta$

– Use polar when the region or integrand involves $x^2 + y^2$, circles, or sectors

Example 3: The Gaussian Integral (I^2 Trick)

Evaluate $I = \int_{-\infty}^{\infty} e^{-x^2} dx$.

1. **Square the integral:** $I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA$

2. **Convert to polar:** $x^2 + y^2 = r^2$, region is all of $\mathbb{R}^2$.

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} \cdot r dr d\theta$$

3. **Inner integral:** Let $u = -r^2$, $du = -2r dr$: $\int_0^{\infty} r e^{-r^2} dr = \left[-\frac{1}{2}e^{-r^2}\right]_0^{\infty} = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$

4. **Outer integral:** $I^2 = 2\pi \cdot \frac{1}{2} = \pi$

5. **Result:** $I = \boxed{\sqrt{\pi}}$

This integral cannot be evaluated by single-variable techniques — the I^2 trick with polar coordinates is essential.

6. Complex Regions (Splitting)

When a region is neither purely Type I nor Type II – **split** it:

$$\iint_R f \, dA = \iint_{R_1} f \, dA + \iint_{R_2} f \, dA$$

You can even use Cartesian for one piece and polar for another.

Example 4: Split Region: Cartesian + Polar

Evaluate $\iint_R (1 + x^2 + y^2) \, dA$ where R is the region inside the square $[-1, 1] \times [-1, 1]$ but outside the unit disk $x^2 + y^2 < 1$, combined with the upper half-disk $x^2 + y^2 \leq 1, y \geq 0$.

1. **Split:** Let R_1 = upper half-disk ($x^2 + y^2 \leq 1, y \geq 0$) and R_2 = square minus full disk (corners).
2. R_1 **in polar:** $0 \leq r \leq 1, 0 \leq \theta \leq \pi$.

$$\iint_{R_1} (1 + r^2) \cdot r \, dr \, d\theta = \int_0^\pi \int_0^1 (r + r^3) \, dr \, d\theta = \pi \left[\frac{r^2}{2} + \frac{r^4}{4} \right]_0^1 = \frac{3\pi}{4}$$

3. R_2 : The corners of the square outside the disk can be set up as a Cartesian integral:

$$\iint_{\text{square}} (1 + x^2 + y^2) \, dA - \iint_{\text{disk}} (1 + x^2 + y^2) \, dA$$

Square integral: $\int_{-1}^1 \int_{-1}^1 (1 + x^2 + y^2) \, dy \, dx = \frac{20}{3}$.

Full disk integral (polar): $\int_0^{2\pi} \int_0^1 (1 + r^2)r \, dr \, d\theta = 2\pi \cdot \frac{3}{4} = \frac{3\pi}{2}$.

Corners: $\frac{20}{3} - \frac{3\pi}{2}$.

4. **Combine:** Total = $\frac{3\pi}{4} + \frac{20}{3} - \frac{3\pi}{2} = \frac{20}{3} - \frac{3\pi}{4}$

5. **Result:** $\frac{20}{3} - \frac{3\pi}{4}$

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^x xy \, dy \, dx$.

Answer: $1/8$

Problem 2. Evaluate $\iint_R xy \, dA$ where R is bounded by $y = 0$, $y = \sqrt{x}$, and $x = 4$. (Type I)

Answer: $32/3$

Problem 3. Evaluate $\iint_D e^{-(x^2+y^2)} dA$ where D is the annulus $1 \leq x^2 + y^2 \leq 4$. (Use polar.)

Answer: $\pi(e^{-1} - e^{-4})$

Problem 4. Reverse the order of integration and evaluate: $\int_0^1 \int_0^{\sqrt{y}} f(x, y) dx dy$ where $f(x, y) = e^{y/x}$.

Answer: $\int_0^1 \int_{x^2}^1 e^{y/x} dy dx$

Problem 5. Evaluate $\iint_R \sqrt{x^2 + y^2} dA$ where R is the region in the first quadrant between $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$.

Answer: $\frac{\pi}{2} \cdot \frac{26}{3} = \frac{13\pi}{3}$